ERIC T. CHANG

eric.chang@columbia.edu ◇ ricandrobots.com

EDUCATION

Columbia University, New York, NY **expected 2026**
Ph.D. in Mechanical Engineering GPA: 4.00/4.00, advisor: Matei Ciocarlie
M.S. in Mechanical Engineering (Fall 2022) GPA: 4.03/4.00
NASA Graduate Research Fellow (NSTGRO)

Duke University, Durham, NC **Spring 2021**
B.S.E. in Mechanical Engineering, B.A. in Computer Science GPA: 3.97/4.00
Magna Cum Laude, Graduation With Distinction

RESEARCH INTERESTS

In my Ph.D., I work on building touch sensors to help robots be more dexterous! I'm most interested in developing compact, multimodal tactile fingers, studying new methods of tactile sensing, and exploring how to process rich, multimodal touch data in the context of manipulation.

AWARDS AND HONORS

Graduate Research Fellowship (NSTGRO), *NASA* **2022 - 2026**
Best Inventions of 2023 (Khandate et al., 2023), *TIME Magazine* **2023**
Oscar and Vera Byron Fellowship, *Columbia Engineering* **2021**
Raymond C. Gaugler Award in Materials Science & Engineering, *Duke Engineering* **2021**
Best Poster Award, *Materials Research Society 2021 Virtual Spring Conference* **2021**
Symposium Award (2nd place), *Materials Research Society 2021 Virtual Spring Conference* **2021**
Tau Beta Pi Engineering Honors Society (Treasurer), *Duke Engineering* **2019 - 2021**
Pi Tau Sigma Mechanical Engineering Honors Society, *Duke Engineering* **2019 - 2021**
Pratt Research Fellowship, *Duke Engineering* **2020**
Dean's Undergraduate Research Fellowship, *Duke Undergraduate Research Support Office* **2020**

INDUSTRY EXPERIENCE

Amazon Robotics, Westborough, MA **Summer & Fall 2025**
Research Scientist II Co-op, Innovation Lab, manager: Taskin Padir, Tye Brady
Project: Explore Fingertip Proximity (ToF) and Tactile Sensing for Robotic Manipulation

- First authored manuscript on the utility of fingertip ToF sensing for manipulation
- Supported tactile development and integration efforts across multiple Amazon teams

Nauticus Robotics (formerly Houston Mechatronics), Houston, TX **Summer 2021**
Robotics R&D Intern, manager: John Yamokoski, Adam Konneker
Project: Refine point cloud compression algorithms for underwater untethered data transmission

- Investigated optimizations for compressing point cloud data from TOF, structured light, and lidar sensors with compression ratio $>$ 300 (C++, Python, ROS, Docker)

Realtime Robotics, Boston, MA **Summer 2020**
Mechanical and Applications Engineering Intern, manager: Nathan Koontz, Ty Tremblay
Project: Develop test cell for application of company's motion planning technology to spot welding cells

- Designed and prototyped scaled spot-welding gun and work cell for testing for a major customer (OnShape)
- Wrote software to control robots and weld guns for bimanual motion planning demo (Python, Arduino)

Coherix, Ann Arbor, MI **Summer 2019**
Product Development Intern
Project: Improve robot programming methods in automotive manufacturing

- Improved potential plant efficiency by 25 minutes per part through development of machine vision software to self-correct manually programmed nozzle position of an adhesive-dispensing robot (Python, FANUC arm)

RESEARCH EXPERIENCE

R.O.A.M. Lab, Columbia University **Fall 2021 - Present**
Ph.D. Candidate, advisor: Matei Ciocarlie

- Designing multimodal tactile fingers for dexterous manipulation
- Took courses in robotics, control, dynamics, mechatronics, machine learning, robot learning
- Research interests: tactile sensing (multimodal tactile finger design, tactile vibration sensing), touch processing, dexterous manipulation

Intelligent Robotics Group, NASA Ames Research Center **Fall 2023, Summer 2024**
NSTGRO Fellowship Program Intern, advisors: Trey Smith, Brian Coltin

- Improved design of 3-fingered underactuated hand for Astrobee (ISS' free-flyer robot)
- Work towards integrating tactile sensor work into NASA applications, e.g. for intravehicular space robots

Mitzi Research Group, Duke University **Spring 2018 - Spring 2021**
Research Assistant, advisor: David B. Mitzi

- First authored paper on bournonite band gap engineering, working to develop solar materials and devices that are cost effective and sustainable

Duke Robotics Club, Duke University **Spring 2018 - Spring 2021**
Task Planning Lead, Mechanical Engineer
Project: Design autonomous underwater vehicle for and compete in International RoboSub Competition

- Designed and implemented task planning architecture (Python); designed, prototyped, and tested servo-controlled torpedo launcher (iterative design, Solidworks)

PUBLICATIONS

Peer-Reviewed Publications [* indicates joint first authorship]

[C.5] **E.T. Chang***, P. Ballentine*, Z. He*, D. Kim, K. Zhang, H. Liang, J. Palacios, W. Wang, P. Piacenza, I. Kymissis, M. Ciocarlie, "SpikeATac: A Multimodal Tactile Finger with Taxelized Dynamic Sensing for Dexterous Manipulation" *ICRA* **2026**. https://arxiv.org/abs/2510.27048

[C.4] K. Zhang*, D. Kim*, **E.T. Chang***, H. Liang, Z. He, K. Lampo, P. Wu, I. Kymissis, M. Ciocarlie, "VibeCheck: Using Active Acoustic Tactile Sensing for Contact-Rich Manipulation," *IROS* **2025**. https://arxiv.org/abs/2504.15535

[C.3] G. Khandate, B. Wang, S. Park, W. Ni, J. Palacios, **E.T. Chang**, P. Wu, R. Ho, M. Ciocarlie, "Train Robots in a JIF: Joint Inverse and Forward Dynamics with Human and Robot Demonstrations," *ArXiV* **2025** https://arxiv.org/abs/2503.12297

[J.3] G. Khandate*, T. Saidi*, S. Shang*, **E.T. Chang**, Y. Liu, S. Dennis, J. Adams, M. Ciocarlie, "R×R: Rapid eXploration for Reinforcement Learning via Sampling-based Reset Distributions and Imitation Pre-training," *Autonomous Robots* **2024** (RSS 2023 special issue). https://arxiv.org/abs/2401.15484

[C.2] **E.T. Chang***, R. Wang*, P. Ballentine, J. Xu, T. Smith, B. Coltin, I. Kymissis, M. Ciocarlie, "An Investigation of Multi-feature Extraction and Super-resolution with Fast Microphone Arrays," *IEEE Intl. Conf. on Robotics and Automation (ICRA)* **2024**. https://arxiv.org/abs/2310.00206

[C.1] G. Khandate*, S. Shang*, **E.T. Chang**, T.L. Saidi, J. Adams, M. Ciocarlie, "Sampling-based Exploration for Reinforcement Learning of Dexterous Manipulation," *Robotics: Science and Systems (RSS)* **2023**. https://arxiv.org/abs/2303.03486

- **Named to TIME's Best Inventions of 2023**

[J.2] **E.T. Chang**, G. Koknat, G.C. McKeown Wessler, Y. Yao, V. Blum, D.B. Mitzi, "Phase Stability, Band Gap Tuning, and Rashba Splitting in Selenium-Alloyed Bournonite: $CuPbSb(S_{1-x}Se_x)_3$," *Chemistry of Materials* **2023** *35*, 595-608. https://doi.org/10.1021/acs.chemmater.2c03109

[J.1] S. Tran, J. Chen, G. Kozel, **E.T. Chang**, et al., "Development of an optically transparent kidney model for laser lithotripsy research," *BJU International* **2023**. https://doi.org/10.1111/bju.16015

Workshop Papers, Posters, and Technical Reports

[P.2] P. Ballentine, **E.T. Chang**, I. Kymissis, M. Ciocarlie, "A Flexible PVDF-based Dynamic Tactile Sensors for Integration into a Multimodal Robot Finger," June 2025. Oral Presentation. Electronics Materials Conference 2025.

[W.2] **E.T. Chang***, P. Ballentine*, I. Kymissis, M. Ciocarlie, "Spike-a-Tac: Development Towards a PVDF-Based Tactile Finger with distributed Vibration Sensing," May 2024. Extended abstract and poster presentation. ICRA 2024 ViTac Workshop.

[W.1] **E.T. Chang**, P. Ballentine, I. Kymissis, M. Ciocarlie, "Towards Development of a Signal-Dense Multimodal Tactile Finger," June 2023. Extended abstract and poster presentation. ICRA 2023 ViTac Workshop.

[P.1] **E.T. Chang**, G. Koknat, V. Blum, D.B. Mitzi, "Synthesis and Characterization of Selenium-Alloyed Bournonite $CuPbSb(S_{1-x}Se_x)_3$: a Prospective Semiconductor for Optoelectronic Applications," March 2021. Poster presentation. Materials Research Society 2021 Virtual Spring Conference.

- **Won best poster award and placed 2$^{\text{nd}}$ in symposium award.**

[TR.1] Duke RoboSub Team, "CTHULHU: The Design and Implementation of the Duke Robotics Club's 2019/2020/2021 RoboSub Competition Entry," RoboSub: San Diego, USA, 2019/2020/2021.
https://robonation.org/app/uploads/sites/4/2019/10/Duke_RS19_TDR.pdf
https://robonation.org/app/uploads/sites/4/2020/08/RS20_TDR_Duke.pdf
https://robonation.org/app/uploads/sites/4/2021/07/RoboSub_2021_Duke_TDR.pdf
Placed 1$^{\text{st}}$ of 54 (2021) and 4$^{\text{th}}$ of 33 (2020) in technical design report portion of competition

TECHNICAL SKILLS

Proficient	Python, ROS, ROS2, Arduino, MATLAB, LaTeX, 3D CAD (Solidworks, Onshape), 3D printing (FDM, SLA)
Some Experience	C, C++, micro-ROS, PCB design (Altium Designer/Circuitmaker), laser cutting, CNC mill, lathe

SERVICE

- Reviewer for Nature Communications, T-RO, ICRA, IROS, BioRob, RSS
- Created and led an arduino activity for Columbia's Girls' Science Day event (Nov 2021, Spring 2022)